

DANIEL SZUREK

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Tuscaloosa, AL, USA

EDUCATION

The University of Alabama

Master of Science in Computer Science

Accelerated Master's Program (AMP)

May 2025 - December 2026

GPA: 3.67

The University of Alabama

Bachelor of Science in Computer Science

Summa Cum Laude, Minor in Mathematics, Honors College, Upsilon Pi Epsilon

August 2022 - May 2026

GPA: 3.87

PROFESSIONAL EXPERIENCE

General Motors

ADAS Software Engineer Intern

Milford, MI, USA

May 2026 - August 2026

- Engineered an agentic LLM-driven automation pipeline in Python using the GitHub API to auto-generate Event Data Recorder (EDR) signal recording definition functions, reducing manual authoring time by 80%
- Modernized a legacy Simulink model untouched for 4 years, developed embedded C code for the EDR, and built data recording infrastructure supporting over 10 ADAS feature teams for safety-critical validation across thousands of miles
- Collaborated with ADAS feature teams to translate recording requirements into automated, scalable software solutions

Tuscaloosa, AL, USA

The University of Alabama EcoCAR

Connected and Automated Vehicle (CAV) Lead

September 2025 - Present

- Architected the full autonomous navigation stack on a pre-production Cadillac Lyriq, integrating Lane Centering Control, Cooperative Adaptive Cruise Control, and V2X connectivity with stock onboard systems, then validated safety-critical performance through Hardware-in-the-Loop (HIL) and Vehicle-in-the-Loop (VIL) testing
- Developed and tuned ADAS control algorithms in MATLAB/Simulink including an MPC/PID longitudinal controller, Stanley lateral controller, and GNN sensor fusion pipeline, cutting lateral tracking error to under 30 cm in HIL testing
- Led a 15-person team in GM's EcoCAR EV Challenge, earning 1st in Vehicle Technical Specifications, 2nd in MathWorks Model-Based Design, and 2nd in CAV Presentations

PROJECTS

Gaussian Process and MPC Controller for Automatic Intersection Navigation via V2X

[Link to Project](#)

- Developed a Gaussian Process-based MPC controller for intersection navigation robust to degraded V2I communication, achieving zero red-light violations across 35 configurations (up to 33% packet loss) and 38% energy savings via GLOSA

Driveability Accessible Driving Resource

[Link to Project](#)

- Built a social-media web platform with resources and reviews on assisted-driving tools for disabled users, completed as my degree capstone project

Super-Resolution Facial Recognition Pipeline

[Link to Project](#)

- Developed a custom CNN pipeline with a Deep Super-Resolution front end and quantized EdgeFace model for facial recognition on low-resolution inputs, improving accuracy while reducing model size for efficient deployment

PUBLICATIONS

2026 IEEE Transportation Electrification Conference & Expo (ITEC+EATS), Novi, MI, USA, Jun. 2026

- D. Szurek, C. Thirumala Setty, C. Hall, and H.-S. Yoon, "Gaussian Process-Based MPC for Robust Autonomous Intersection Navigation Under Degraded V2I Communication."
 - Link: <https://ieeexplore.ieee.org/document/11593068>
- C. Hall, C. Thirumala Setty, D. Szurek, et al., "Lane Centering Under Camera Failures: Classical Control vs. Reinforcement Learning for ADAS."
 - Link: <https://ieeexplore.ieee.org/document/11592966>
- C. Thirumala Setty, D. Szurek, et al., "Adaptive MPC Weight Tuning via Reinforcement Learning for Eco-Driving: Framework and Oracle Gap Analysis."
 - Link: <https://ieeexplore.ieee.org/document/11592971>

TECHNICAL SKILLS

ADAS & Autonomy: ROS 2, Simulink, RTMaps, Model Predictive Control (MPC), Stanley Controller, Sensor Fusion (GNN), Hardware-in-the-Loop (HIL), Vehicle-in-the-Loop (VIL), V2X / DSRC, dSPACE AUTERA, embedded software programming, CAN

AI & ML: PyTorch, Computer Vision (CNNs), Gaussian Processes, Reinforcement Learning, LLM Integration (OpenAI API), LangChain

Cloud & DevOps: Azure (Data Factory, Databricks), Docker, Git, Linux (Ubuntu)

Languages: Python, C/C++, MATLAB, SQL, Bash, Flutter, JavaScript, Swift, TypeScript, HTML